

AUTOMATIC FACE RECOGNITION SYSTEM USING PRINCIPAL COMPONENT ANALYSIS AND EIGENFACE APPROACH

Guttula Lakshmi Satya Lavanya^{*1}

^{*1}Student, Department Of Electronics And Communication Engineering, Andhra University
College Of Engineering, Visakhapatnam, India.

DOI : <https://www.doi.org/10.56726/IRJMETS30006>

ABSTRACT

Face Recognition is one of the most important techniques to authenticate a person in a secure system, for criminal identification, for passport verification, terrorist detection and so on. In our methodology, to recognize a human face we use approach known as Eigenface method and to reduce the dimensionality of the original image we use the technique known as Principal Component Analysis. PCA is a statistical approach used for reducing the number of variables in face recognition. In our proposed methodology the accuracy for the recognition of a face is increased compared to the previous methods and also changes to the background illuminations can be overcome. The original image can be reconstructed from the eigen faces to have a better resolution while displaying. In this we are going to study and understand the Eigenface method by reducing the dimensions of the original image. The Experimental results shows the accuracy of recognition of the test face with the trained data base by applying our proposed methodology.

Keywords: Face Recognition, Principal Component Analysis, Eigen Face, Eigen Vectors, Covariance.

I. INTRODUCTION

Over the last ten years, face recognition has become a one of the popular areas of research in computer vision and one of the most successful applications of image analysis and understanding. Due to the nature of the problem, not only in the computer science also occurs in neuroscientists and psychologists. This is the general opinion that advances in computer vision research will provide useful thoughts to neuroscientists and psychologists into how human brain works, and vice versa [1]. The goal is to implement the system for a face and distinguish it from a large number of stored faces with some real-time variations. It gives the efficient way to find the lower dimensional space. Further this algorithm can be implemented to recognize the gender of a person or to interpret the facial expression of a person. Recognition could be carried out under widely varying conditions like frontal view, scaled frontal view etc. are tried, while the training data set covers limited views. The aim of this research paper is to study and develop an efficient method for face recognition using Eigen face approach with principal component analysis and to perform test for program optimization and accuracy. This approach is preferred due to its simplicity, speed and learning capability.

II. METHODOLOGY

Face Recognition

One of the simplest and the most effective PCA approach used in the face recognition systems called eigenface approach. This approach transforms the faces into the small set of essential characteristics like eigenfaces, these are the main components of the initial set of images i.e., training set. The Recognition is done by projecting the new image in the eigenface subspace after which person is classified by comparing its position in the eigenface space with the position of known individuals. The main advantage of this approach over other face recognition systems is in its simplicity, speed and insensitivity to small or gradual changes on the face.

The entire recognition process includes two steps:

1. Initialization process.
2. Recognition process

The Initialization process involves the below procedure:

1. Obtain the face images known as training data.
2. Calculate the Eigenfaces from the training data use only the high value of eigenvalues. The M images describes the face space.

3. Calculate distribution in the M-dimensional space for individual known person by projecting face images onto the face-space.
4. Obtain the weights depends up on the input image and the M eigenfaces by transforming the input image into the individual Eigenfaces.
5. Calculate the image if it is known or unknown by checking to see if the image is closed matched to the free space.
6. If it is a face, then find the feature weight to the known person or unknown.
7. Update the eigenfaces or weights as either a known or unknown, if the same face is repeated more times, then again calculate the feature weight.

III. PROPOSED MODELING AND ANALYSIS

Proposed method:

The proposed Methodology integrates both initialization and recognition of the face. These are the fundamental steps to match the unknown face with the set of data images in the data set. The following block diagram indicates the various steps involved in the algorithm.

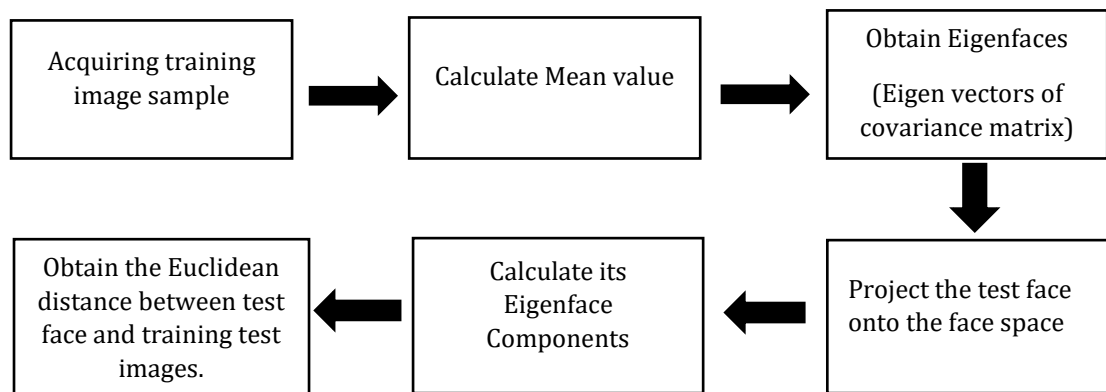


Figure 1: Block Diagram for Proposed Methodology

Let a face image $\Gamma(x, y)$ be a two-dimensional M by N array of intensity values. An image may also be considered as a vector of dimension M $\times$ N. The following are the steps involved in our methodology.

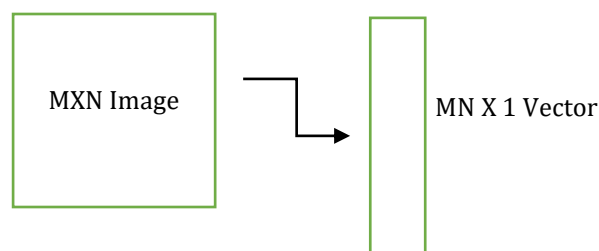


Figure 2: Conversion of MXN Image into MN X 1 vector

Step1: Prepare the training data face by Obtaining the face images as I1, I2, I3, I4, IM (training data faces). The face images must be of centered.

Step 2: Obtain the data set with each face image in the data set is converted into a vector and placed into the training data set S.

$$S = \{r_1 \ r_2 \ r_3 \dots r_m\} \dots\dots\dots \text{eq.4.1}$$

Step 3: Calculate the mean vector. The mean vector is also known as Average value and it is denoted as (Ψ). It is calculated by using the formula:

$$\Psi = \frac{1}{M} \sum_{n=1}^{\bar{M}} \tau_n \dots\dots\dots \text{eq.4.2}$$

Step 4: Take the difference value between the original face images and the mean value. Store the obtained subtracted value in the variable ϕ_i .

$$\phi_i = \tau_i - \Psi \dots\dots\dots \text{eq.4.}$$

Step 5: Calculate the covariance matrix with the help of following formula.

$$C = \frac{1}{M} \sum_{n=1}^M \Phi_n \Phi_n^T \dots\dots\dots \text{eq.4.4}$$

$$= AA^T \quad (N^2 \times N^2 \text{ Matrix Where, } A = \{\Phi_1, \Phi_2, \dots, \Phi_M\})$$

Step 6: Obtain the eigenvalues and eigen vectors of the covariance matrix. The obtained covariance matrix has dimensionality of $N^2 \times N^2$, so it has a N^2 eigen face and eigen values.

Step 7: Consider only K eigenvectors which have highest values compared to the other values. Eigen faces with lowest eigen values can be eliminated because these are the small features of the face.

Step 8: The feature weight for training data base images can be calculated by using the formula

$$\omega_i = u_i^T (\tau_i - \psi) \dots\dots\dots \text{eq.4.5}$$

Where U_i is the i^{th} Eigenfaces and $i=1, 2, 3, \dots, K$. The obtained weight can be defined as a vector as shown below.

$$\Omega^T I = (\omega_1, \omega_2, \omega_3, \dots, \omega_I)$$

To Know the given face is recognized or not the following steps need to be followed.

Step 9: Read the input image which is need to be recognized and separate the face from it.

Step 10: Calculate the feature vector of the input face. The test image is transformed into its eigenface components. First, we compare line of our input image with our mean image and multiply their difference with each eigenvector. Each value would represent a weight and it would be defined by a vector.

$$\omega_{\text{test}} = u_i^T (\tau_{\text{test}} - \psi) \dots\dots\dots \text{eq.4.6}$$

Where ' i ' is the i^{th} Eigenface and $i=1, 2, 3, \dots, K$.

Step 11: Calculate the Euclidean distance which is the average distance between test feature vector and all the feature vectors in the training data set.

$$\Omega_{\text{test}}^T = [\omega_1, \omega_2, \dots, \omega_K]$$

Step 12: Mathematically, the minimum Euclidean distance can be calculated by using the below equation

$$\epsilon_k = \sqrt{||\Omega_{\text{test}} - \Omega_i||^2} \dots\dots\dots \text{eq.4.7}$$

IV. RESULTS AND DISCUSSION

The computational results of our proposed methodology as shown below figures. Here both the test image and equivalent image which is stored in data base both are matched with the highest possible accuracy.

CASE:1 The test image and the trained data image are matched.

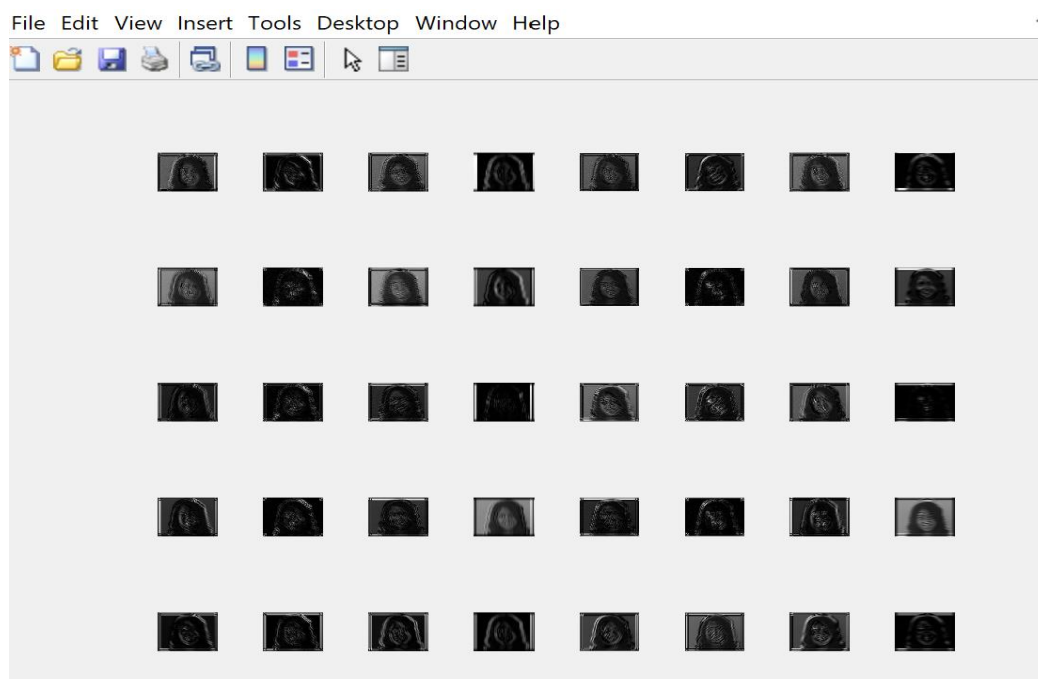


Figure 3: Eigen faces for the input test image.

Here the test image is given which is verified with the images in the trained data base. Here the outcome is that the given face is matched with the trained data base. The result which is shown in below figure 4.

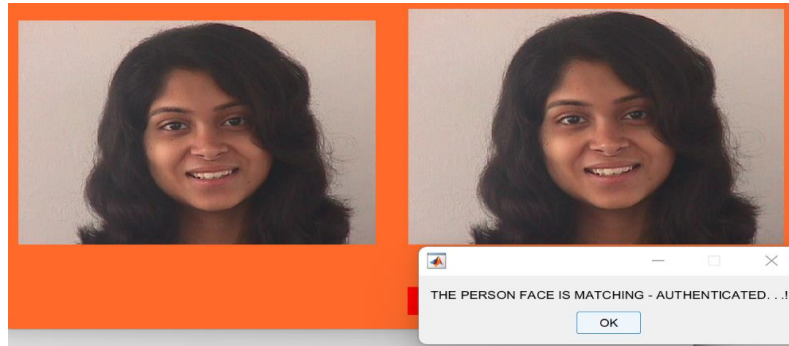


Figure 4: The test image is recognized.

CASE:2 The test image and the trained data image are NOT matched.

Some of the highest valued eigen vectors are considered among all the eigenfaces. Some of them are shown in below figure 5.

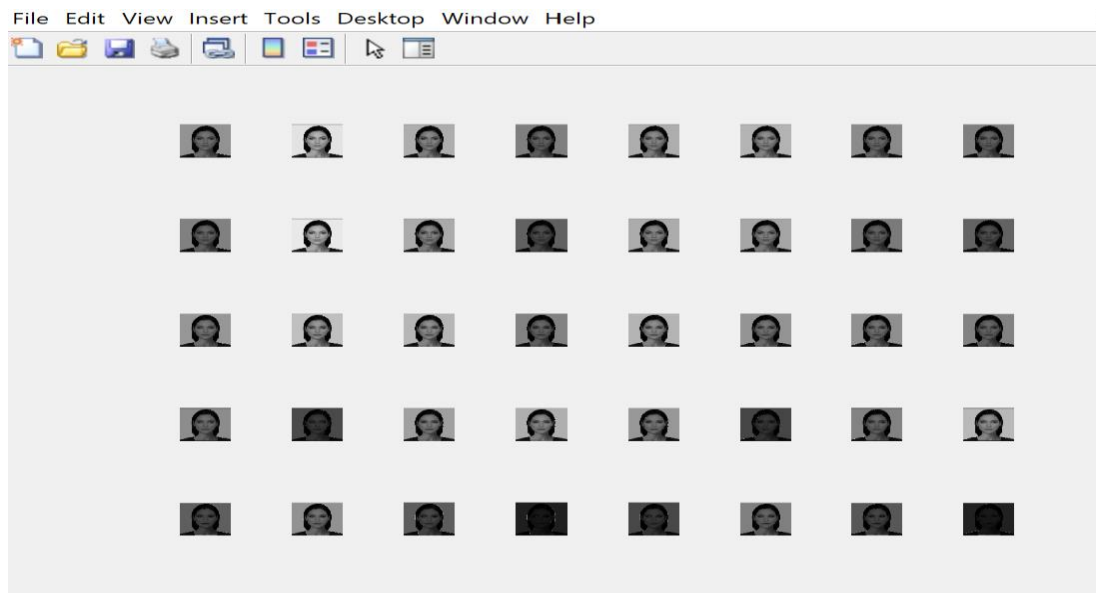


Figure 5: Eigen faces for the input test image.

Here the test image is given which is verified with the images in the trained data base. Here the outcome is that the given face is NOT matched with the trained data base. The result which is shown in below figure 6.

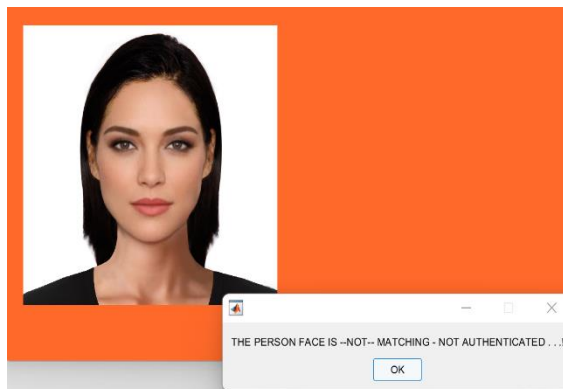


Figure 6: The test image is recognized

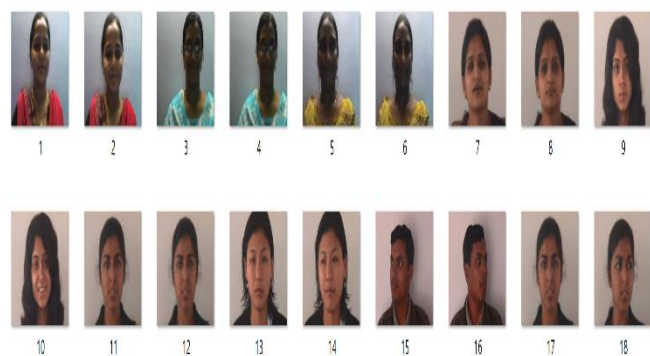


Figure 7: The trained data images.

Here given input image which is in figure 6 is not matched with the trained data base which is shown in figure 7. Hence the test face is not authenticated.

V. CONCLUSION

In order to improve the performance in face detection, a technique known as Eigen face approach was proposed in this work. The proposed method combines Principal Component Analysis (PCA), results in reducing the dimensional vectors in the image.

In this project, Principal component analysis approach to the face recognition problem was studied and a face recognition system based on the eigenfaces approach was proposed. The algorithm has been tested for the image database implemented using MATLAB.

The algorithm developed in a generalized one which works well with any type of images. The tests conducted on PNG images and JPEG images. The eigenface approach thus provides a practical solution that is well fitted to the problem of face recognition. It is fast, relatively simple and has been shown to work well in constrained environment.

VI. REFERENCES

- [1] M. A. Turk and A. P. Pentland, "Face Recognition Using Eigenfaces," in Proc. IEEE Conference on Computer Vision and Pattern Recognition, pp. 586–591. 1991.
- [2] Nirmalya Kar, Mrinal Kanti Debbarma, Ashim Saha, and Dwijen Rudra Pal "Study of Implementing Automated Attendance System Using Face Recognition Technique "International Journal of Computer and Communication Engineering, Vol. 1, No. 2, July 2012.
- [3] Goldstein, A. J., Harmon, L. D., and Lesk, A. B., "Identification of human faces", Proc. IEEE 59, pp. 748-760, (1971).
- [4] M.N. Shah Zainudin., Radi H.R., S. Muniroh Abdullah., Rosman Abd. Rahim., M. Muzafar Ismail," Face Recognition using PCA and LDA", International Journal of Electrical & Computer Sciences.
- [5] Yohei KAWAGUCHI, Tetsuo SHOJI, Weijane LIN, and Koh KAKUSHO, "Face Recognition-based Lecture Attendance System", Department of Intelligence Science and Technology, Graduate School of Informatics, Kyoto University.
- [6] L. Stanca, "The Effects of Attendance on Academic Performance: Panel Data Evidence for Introductory Microeconomics," J. Econ. Educ., vol. 37, no. 3, pp. 251–266, 2006.
- [7] P.K. Pani and P. Kishore, "Absenteeism and performance in a quantitative module A quantile regression analysis," Journal of Applied Research in Higher Education, vol. 8 no. 3, pp. 376-389, 2016.
- [8] U. Thakar, A. Tiwari, and S. Varma, "On Composition of SOAP Based and RESTful Services," IEEE 6th Int. Conference on Advanced Computing (IACC), 2016.
- [9] K.P.M. Basheer and C.V. Raghu, "Fingerprint attendance system for classroom needs," Annual IEEE India Conference (INDICON), pp. 433-438, 2012.
- [10] S. Konatham, B.S. Chalasani, N. Kulkarni, and T.E. Taeib, "Attendance generating system using RFID and GSM," IEEE Long Island Systems, Applications and Technology Conference (LISAT), 2016.
- [11] S. Noguchi, M. Niibori, E. Zhou, and M. Kamada, "Student Attendance Management System with Bluetooth Low Energy Beacon and Android Devices," 18th International Conference on Network-Based Information Systems, pp. 710-713, 2015.
- [12] S. Chintalapati and M.V. Raghunadh, "Automated attendance management system based on face recognition algorithms," IEEE Int. Conference on Computational Intelligence and Computing Research, 2013.
- [13] G. Wang, "Improving Data Transmission in Web Applications via the Translation between XML and JSON," Third Int. Conference on Communications and Mobile Computing (CMC), pp. 182-185, 2011.
- [14] X. Zhu, D. Ren, Z. Jing, L. Yan, and S. Lei, "Comparative Research of the Common Face Detection Methods," 2nd International Conference on Computer Science and Network Technology, pp. 1528-1533, 2012.
- [15] V. Gupta and D. Sharma, "A Study of Various Face Detection Methods," International Journal of Advanced Research in Computer and Communication Engineering vol. 3, no. 5, pp.6694-6697, 2014.
- [16] P. Viola and M.J. Jones, "Robust Real-Time Face Detection," International Journal of Computer Vision, vol. 57, no. 2, pp. 137-154, 2004.